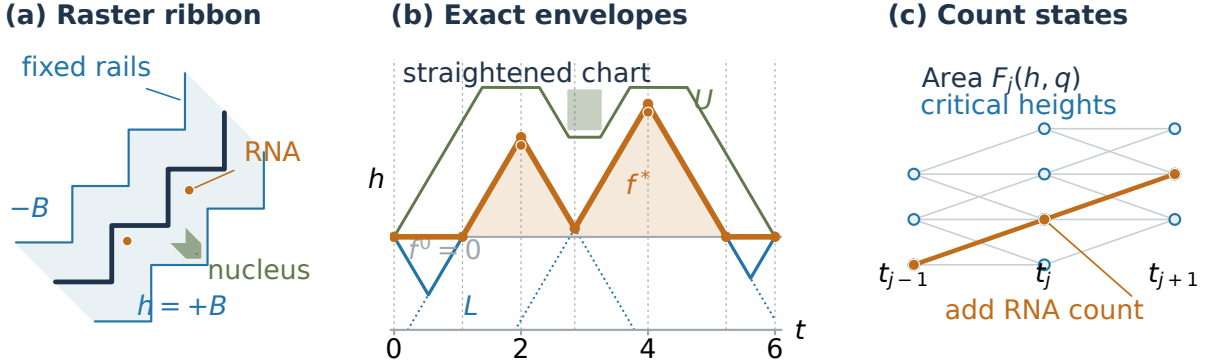

(a) Raster ribbon
fixed rails
RNA
−B
nucleus
h = +B
(b) Exact envelopes
straightened chart
U
h
f*
f⁰ = 0
L
0
2
4
6
t
(c) Count states
Area F_j(h, q)
critical heights
t_j − 1
t_j
t_j + 1
add RNA count
RNA constraints ⟶ exact envelopes ⟶ minimum area by count